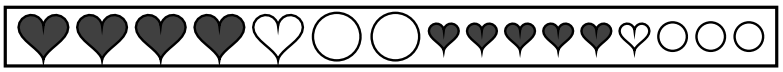
Preview: set-01/by_level/Grade_03-04_Ecolier/y2024_q24 (v1)

Taking turns: what is still left depends on who went first - try the orders and keep track

Part A: Each question is worth 3 points

1. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Emma takes all the **heart-shaped** cookies that are still on the tray.
Rosa takes all the **white** cookies that are still on the tray.
Tina takes all the **large** cookies that are still on the tray.
First Rosa goes, then Tina, then Emma. How many cookies does **Emma** take?



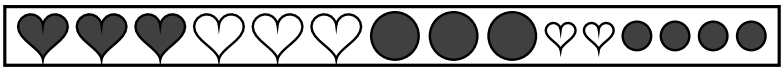
A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 8 ***E.*** 11

2. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Mia takes all the **heart-shaped** cookies that are still on the tray.
Zoe takes all the **white** cookies that are still on the tray.
Nora takes all the **large** cookies that are still on the tray.
First Zoe goes, then Mia, then Nora. How many cookies does **Nora** take?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 5 ***E.*** 11

Part B: Each question is worth 4 points

3. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Mia takes all the **heart-shaped** cookies that are still on the tray.
Rosa takes all the **white** cookies that are still on the tray.
Zoe takes all the **large** cookies that are still on the tray.
We do not know in which order they go. How many cookies are still on the tray after all three girls have taken theirs?

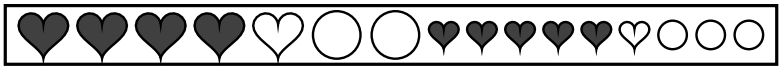


A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

4. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Rosa takes all the **heart-shaped** cookies that are still on the tray.
Emma takes all the **white** cookies that are still on the tray.
Lena takes all the **large** cookies that are still on the tray.
We do not know in which order they go. How many cookies are still on the tray after all three girls have taken theirs?

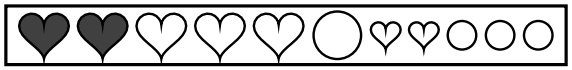
A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

5. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Ida takes all the **heart-shaped** cookies that are still on the tray.
Mia takes all the **white** cookies that are still on the tray.
Lena takes all the **large** cookies that are still on the tray.
One of the girls takes 2 cookies, one takes 3 and one takes 11. In which order did the girls take their cookies?



A. Mia, Lena and then Ida ***B.*** Mia, Ida and then Lena ***C.*** Lena, Ida and then Mia
D. Lena, Mia and then Ida ***E.*** Ida, Lena and then Mia

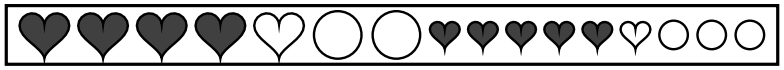
6. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Lena takes all the **heart-shaped** cookies that are still on the tray.
Tina takes all the **white** cookies that are still on the tray.
Emma takes all the **large** cookies that are still on the tray.
One of the girls takes 2 cookies, one takes 3 and one takes 6. In which order did the girls take their cookies?



A. Emma, Tina and then Lena ***B.*** Tina, Lena and then Emma ***C.*** Lena, Tina and then Emma
D. Emma, Lena and then Tina ***E.*** Tina, Emma and then Lena

Part C: Each question is worth 5 points

7. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Lena takes all the **heart-shaped** cookies that are still on the tray.
Mia takes all the **white** cookies that are still on the tray.
Rosa takes all the **large** cookies that are still on the tray.
One of the girls takes 2 cookies, one takes 3 and one takes 11. How many cookies did **Mia** take?

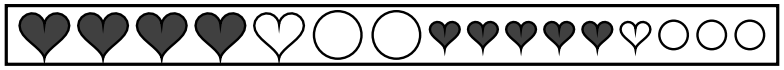


A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 7

8. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Mia takes all the **heart-shaped** cookies that are still on the tray.
Tina takes all the **white** cookies that are still on the tray.
Rosa takes all the **large** cookies that are still on the tray.
One of the girls takes 2 cookies, one takes 3 and one takes 8. How many cookies did **Mia** take?

A. 7 ***B.*** 8 ***C.*** 9 ***D.*** 10 ***E.*** 11

9. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Mia takes all the **heart-shaped** cookies that are still on the tray.
Lena takes all the **white** cookies that are still on the tray.
Nora takes all the **large** cookies that are still on the tray.
One of the girls takes 2 cookies, one takes 3 and one takes 11. Which of these groups of cookies did one of the girls take?



A. ***B.*** ***C.*** ***D.*** ***E.***

10. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Ida takes all the **heart-shaped** cookies that are still on the tray.
Zoe takes all the **white** cookies that are still on the tray.
Nora takes all the **large** cookies that are still on the tray.
One of the girls takes 3 cookies, one takes 5 and one takes 6. Which of these groups of cookies did one of the girls take?

A. ***B.*** ***C.*** ***D.*** ***E.***

Answer Key

1.	**A**	**2.**	**A**	**3.**	**E**	**4.**	**E**	**5.**	**E**	**6.**	**D**
7.	**C**	**8.**	**B**	**9.**	**A**	**10.**	**A**				

Solutions

1. A (5) — Cross out what the girls before Emma take first. Of her cookies only 5 are still on the tray (she would get 11 if she went first). [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · given_order]

2. A (1) — Cross out what the girls before Nora take first. Of her cookies only 1 are still on the tray (she would get 5 if she went first). [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · given_order]

3. E (4) — A cookie stays only if it is not a heart, not white and not large: the small dark round ones. There are 4 of them - the order does not matter. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · left_over]

4. E (4) — A cookie stays only if it is not a heart, not white and not large: the small dark round ones. There are 4 of them - the order does not matter. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · left_over]

5. E (Ida, Lena and then Mia) — Try the orders and count what is still on the tray. Only Ida, Lena and then Mia gives Ida 11, Lena 2, Mia 3. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · which_order]

6. D (Emma, Lena and then Tina) — Try the orders and count what is still on the tray. Only Emma, Lena and then Tina gives Emma 6, Lena 2, Tina 3. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · which_order]

7. C (3) — Try the orders one by one. Only Lena, Rosa and then Mia gives 2, 3 and 11. So Mia took 3. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · how_many_one]

8. B (8) — Try the orders one by one. Only Mia, Tina and then Rosa gives 2, 3 and 8. So Mia took 8. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · how_many_one]

9. A — Try the orders. Only Mia, Nora and then Lena gives 2, 3 and 11: Mia takes 11, Nora takes 2, Lena takes 3. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · which_group]

10. A — Try the orders. Only Ida, Zoe and then Nora gives 3, 5 and 6: Ida takes 5, Zoe takes 6, Nora takes 3. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · which_group]